CAREPLUS AI

Smart Patient Queue & Virtual Waiting System for Single-Doctor Clinics

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Introduction

Small & Single-Doctor Clinics
Long waits, token gives only a number

CAREPLUS AI
Smart Queue & Virtual Waiting System

Know when to come
"Don't wait in the queue"

KEY OFFERINGS

1
Smart Digital Tokens
QR code scan check-in

2
AI Wait Prediction
Dynamic arrival time updates

3
Virtual Waiting
Step away until turn is near

4
Emergency Priority
Automatic queue recalculation

5
Doctor Dashboard
Automated patient notifications

TECHNOLOGY STACK

Frontend	Backend	Database	AI / ML	Notifications
Flutter / React	Python FastAPI	Firebase / MySQL	Scikit-learn	WhatsApp / SMS

Introduction :Motivation

Small & single-doctor clinics rely on traditional token systems

Hours of waiting
Patients wait for hours in local clinics

Number only
A token gives no accurate estimate of consultation time

Variable timings
Consultation time differs per patient, so the schedule is unpredictable

Emergencies
Unexpected emergencies disrupt the schedule

Constant patient-to-staff inquiries
Severe crowding, confusion, and time uncertainty

Introduction :Motivation

Traditional Queue

Predictive, Adaptive Virtual Waiting System
"Don't wait in the queue. Know when to come."

Virtual waiting
Patients step away and return only when their turn is near

Automatic updates
Waiting time updates when the queue changes or an emergency arrives

Low-cost & portable
A simple setup suited to small clinics

Literature Review

Ref. No.	Paper Details	Findings
[1]	Token Management System M. Sabin T. T. et al., IJARSCT, vol. 2, no. 6, pp. 901-902, Jun. 2022.	• Token-based system for managing patient turns • Gives only a token number, with no accurate consultation time estimate
[2]	Automated Queue Management System M. N. Uddin et al., Global Journal of Management and Business Research, vol. 16, no. 1, pp. 1-8, 2016.	• Automates queue handling in place of manual calling • Does not predict waiting time using AI
-	Various AI patient flow research studies	• AI can support patient flow and waiting time prediction

Literature Review : Observations

Traditional token systems

1
No time estimate
Tokens only issue a number, with no accurate estimate of consultation time

2
Unreliable estimates
Variable consultation times per patient make fixed estimates unreliable

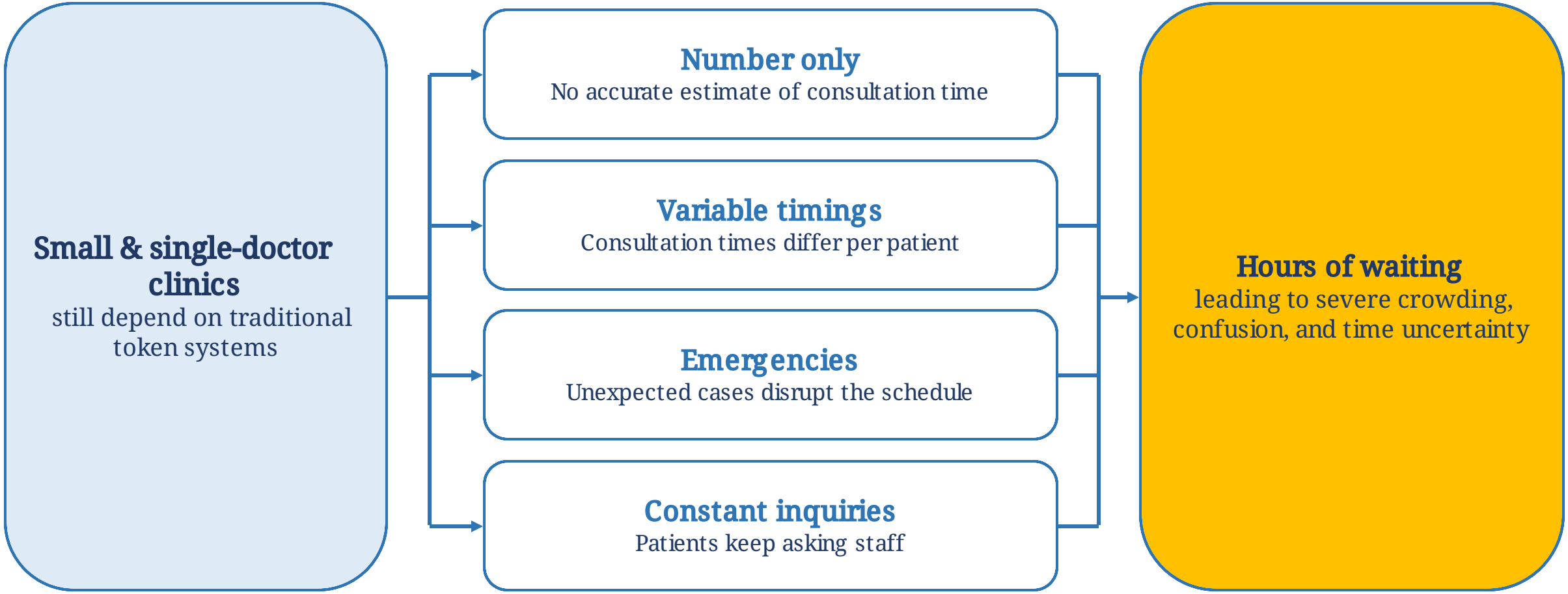
3
Disrupted queue
Unexpected emergencies disrupt the queue and the schedule

4
Crowding
Patients must stay in the clinic, causing crowding and constant inquiries to staff

Literature Review : Gap Analysis

GAP IN TRADITIONAL SYSTEMS	CAREPLUS AI
No accurate waiting time Traditional tokens give only a number	**AI-based waiting time prediction**
Physical waiting Patients sit in a crowded clinic	**Virtual waiting**
Emergencies They disrupt the schedule	**Emergency priority with automatic queue recalculation**
Constant inquiries Patients keep asking staff	**Real-time updates and automated notifications**

Problem Statement



Objectives of Proposed Work

1. Provide smart digital tokens through a QR code scan
2. Give AI-based waiting time predictions with dynamic arrival time updates
3. Enable virtual waiting so patients can step away until their turn is near
4. Handle emergencies with priority and automated queue recalculation
5. Provide a simple doctor dashboard and automated patient notifications

Proposed Work :Block Diagram

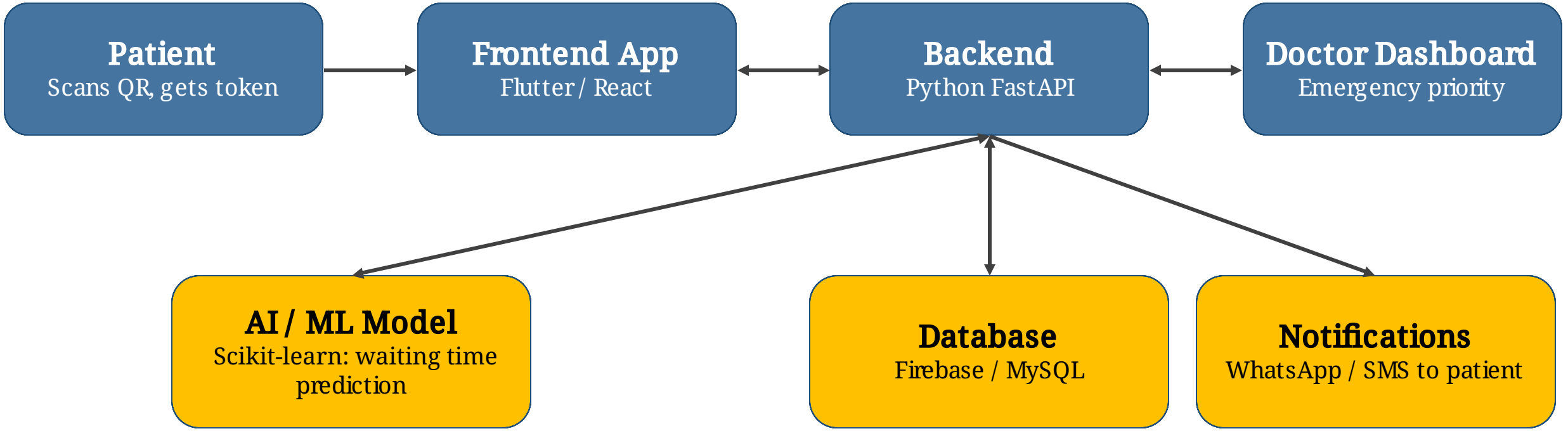


Fig 1.1: Block diagram of proposed work

Proposed Work :Work Flow

1
Check-in
Patient scans the clinic's QR code and receives a digital token

2
Prediction
System estimates the approximate consultation time

3
Virtual Waiting
Patient leaves or waits elsewhere instead of a crowded clinic

4
Dynamic Updates
Estimated time updates automatically in real time when the queue changes

5
Notification
Patient is notified when their turn is near and returns to check in

6
Consultation & Emergencies
Doctor consults; "emergency priority" triggers automatic queue recalculation

Proposed Work
Hardware & Software Resource

SOFTWARE

Frontend
Flutter or React

Backend
Python FastAPI

Database
Firebase or MySQL

AI / ML
Python and Scikit-learn (waiting time prediction)

Notifications
WhatsApp / SMS integration

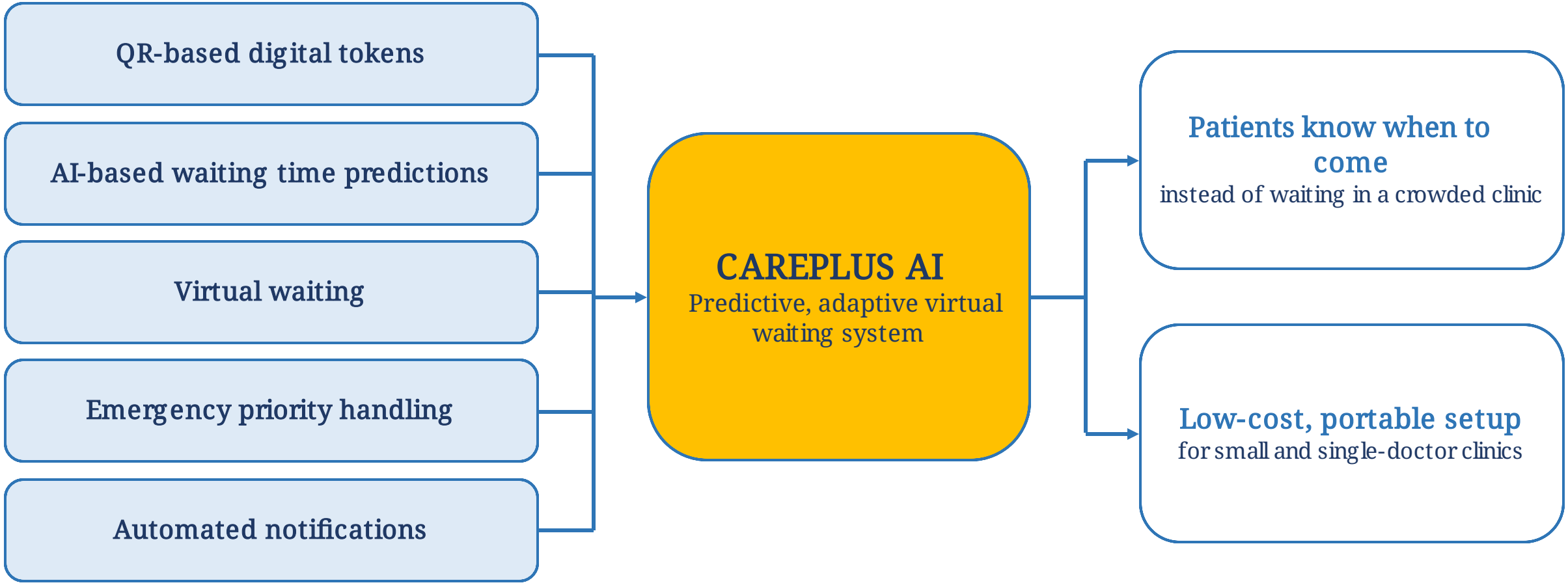
HARDWARE: LOW-COST, PORTABLE SETUP

Android smartphone or tablet

QR codes

Doctor's computer or tablet

Conclusive Discussion



References

JOURNAL ARTICLES

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   M. Sabin T. T. et al., IJARSCT, vol. 2, no. 6, pp. 901-902, Jun. 2022
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Thank You…!!!